Practical: Creating an EC2 Instance and Setting up Apache Web Server

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Aim

To create a virtual server using Amazon Web Services EC2 and configure an Apache Web Server to host a web page.

Theory (Basic Understanding)

- EC2 (Elastic Compute Cloud):
  A service provided by AWS that allows users to launch virtual servers (instances) in the cloud.
- Apache Web Server (httpd):
  A widely used open-source software used to host websites.
- Security Group:
  Acts as a firewall that controls incoming and outgoing traffic to the instance.
- Key Pair:
  A secure login method used to connect to the EC2 instance.

Prerequisites

- AWS Account (Free Tier)
- Internet Connection
- Basic browser knowledge

Step 1: Login to AWS

Open:

[AWS Console]

Login using your AWS account.

Step 2: Open EC2 Service

In AWS search bar type:

EC2

Click:

EC2

Step 3: Launch EC2 Instance

Click:

Launch Instance

Step 4: Configure EC2 Instance

4.1 Name

Enter:

ApacheServer

4.2 Select Ubuntu Image

Choose:

Ubuntu Server 24.04 LTS

(Free Tier Eligible)

4.3 Choose Instance Type

Select:

T2.micro OR t3.micro

(Free Tier Eligible)

4.4 Create Key Pair

Click:

Create new key pair

Enter:

apache-key

Choose:

- Type = RSA
- Format = .pem

Click:

Create key pair

`.pem` file will download automatically.

4.5 Configure Network Settings

Click:

Edit

Enable:

✅ Allow SSH traffic
✅ Allow HTTP traffic

Step 5: Launch Instance

Click:

Launch Instance

Wait until instance state becomes:

Running

Step 6: Connect to EC2 Instance

6.1 Select Instance

Tick/select your instance.

Click:

Connect

6.2 Use EC2 Instance Connect

Choose tab:

EC2 Instance Connect

Click:

Connect

A browser terminal will open.

Step 7: Update Ubuntu Packages

Run:

sudo apt update

Wait for completion.

Step 8: Install Apache Web Server

Run:

sudo apt install apache2 -y

Wait for installation to complete.

Step 9: Start Apache Service

Run:

sudo systemctl start apache2

Step 10: Enable Apache Service

Run:

sudo systemctl enable apache2

Purpose:
Apache automatically starts whenever EC2 restarts.

Step 11: Verify Apache Installation

Go to EC2 Dashboard.

Copy:

Public IPv4 Address

Example:

13.xx.xx.xx

Open browser:

http://PUBLIC_IP

Example:

http://13.xx.xx.xx

You should see:

Apache2 Ubuntu Default Page

This confirms Apache server is running successfully.

Step 12: Create Custom Web Page

Go back to EC2 terminal.

Run:

cd /var/www/html

Step 13: Edit index.html File

Run:

sudo nano index.html

Delete existing content.

Paste:

```
<html>
<head>
<title>My Website</title>
</head>

<body>
<h1>Hello from AWS EC2 🚀</h1>
<p>Apache Web Server Successfully Configured</p>
</body>
</html>
```

Step 14: Save File

Press:

CTRL + X

Then press:

Y

Then press:

Enter

Step 15: Refresh Browser

Refresh:

http://PUBLIC_IP

Your custom webpage will appear.

Common Errors & Solutions

Problem	Reason	Solution
Website not opening	HTTP not enabled	Allow port 80
Connection failed	Instance stopped	Start instance
Blank page	Apache not running	Start httpd
Permission error	Wrong commands	Use sudo

Viva Questions

Q1. What is EC2?
EC2 is a cloud service that provides virtual servers.

Q2. What is Apache?
Apache is a web server used to host websites.

Q3. What is a Security Group?
A firewall that controls traffic.

Q4. What is Key Pair?
Used for secure login to EC2.

Q5. What is the default Apache folder in Ubuntu?
/var/www/html

Q6. Which port does apache use?
Port 80

Q7. Why allow HTTP traffic in Security Group?
To allow users to access the website publicly through browser.